

# Navin Sridhar

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| CONTACT INFO                   | Department of Physics and KIPAC,<br>452 Lomita Mall, Stanford, CA 94305<br>E-mail: <a href="mailto:nsridhar@stanford.edu">nsridhar@stanford.edu</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
| EMPLOYMENT                     | <b>Stanford University, CA</b> <ul style="list-style-type: none"><li>• Simons Collaboration (SCEECs) postdoctoral fellowship/KIPAC (2024–2028)</li><li>• Supervisor: Dr. Roger Blandford</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
| EDUCATION                      | <b>Columbia University, NY</b> <ul style="list-style-type: none"><li>• Ph.D. in Astrophysics (May 2024)</li><li>• M.A. (2020), M.Phil (2021) in Astrophysics<br/>Advisors: Dr. Lorenzo Sironi, Dr. Brian D. Metzger<br/>Thesis: <a href="#">Electromagnetic and Multimessenger Signals From Magnetized Outflows of Compact Object Binaries</a></li></ul> <b>Indian Institute of Science Education &amp; Research, Bhopal</b> <ul style="list-style-type: none"><li>• BS-MS in Physics (CPI: 9.05/10) (May 2018)<br/>Thesis: Accretion disk evolution around black holes, and thermonuclear bursts in neutron stars. (Caltech and TIFR)</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |
| AWARDS AND FELLOWSHIPS         | <ol style="list-style-type: none"><li>1. PI of NASA NICER grant (US\$45,000) (2026–2027)</li><li>2. Neil Gehrels Prize Fellowship: JSI, University of Maryland &amp; NASA (2024)</li><li>3. Brinson Fellowship: Northwestern University/CIERA &amp; University of Chicago (2024)</li><li>4. NASA <i>FINESST</i> award (US\$100,000) (2022–2024)</li><li>5. PI of NASA NICER grant (US\$37,500) (2022–2024)</li><li>6. Simons Foundation Aspen center for physics grant (US\$1,350) (July 2023)</li><li>7. International Astronomical Union (S378) travel grant (US\$1,600) (March 2023)</li><li>8. Columbia Arts and Sciences Graduate Council travel grant (US\$500) (2022)</li><li>9. Columbia ASGC Student/Diversity Initiative Grant (US\$2,000) (2022–2023)</li><li>10. AAS National Osterbrock Leadership fellow (US\$6,400) (2021–2023)</li><li>11. The Yeh Family endowment fellowship, Columbia University (2019–2020)</li><li>12. Dean’s fellowship for graduate study at Columbia University (2018–2020)</li><li>13. Caltech Summer Undergraduate Research Fellowship (US\$6,600) (July–Oct 2017)</li><li>14. Mitacs Globalink Research Internship Award (~CA\$6,600) (May–July 2017)</li><li>15. <i>INSPIRE</i> fellowship by DST, Govt. of India (INR 400,000) (2013–2018)</li></ol>                                                                                                               |  |
| RESEARCH EXPERIENCE/<br>VISITS | <ul style="list-style-type: none"><li>• Affiliated scientist, <a href="#">SCEECs</a> (2023–2027)</li><li>• <b>Simons</b> Collaboration on Extreme Electrodynamics of Compact Sources</li><li>• Visiting Researcher, <b>KITP</b>, UC Santa Barbara (Sept 2024)</li><li>• Visiting Student Researcher, <b>Caltech</b>, Pasadena (2023–2024)<br/>Host: Dr. Sterl Phinney</li><li>• Stellar Interactions and the Transients they Cause, <b>Aspen</b> (July 2023)<br/>Host: Aspen Center for Physics, Colorado</li><li>• Visiting researcher, Simons Foundation – plasma astrophysics program (2019–2023)<br/>Center for Computational Astrophysics, <b>Flatiron Institute</b>, NY<br/>Advisor(s): Dr. Sasha Philippov, Dr. Lorenzo Sironi</li><li>• Summer Undergraduate Research Fellow (SURF), <b>Caltech</b>, Pasadena (July–Oct 2017)<br/>Advisor(s): Dr. Javier García, Prof. Fiona Harrison</li><li>• Mitacs Globalink Research Intern, <b>University of Alberta</b>, Edmonton (May–July 2017)<br/>Advisor: Dr. Rodrigo Fernández</li><li>• Visiting Summer Research Program, <b>TIFR</b>, Mumbai (May–July 2016)<br/>Advisor: Prof. Sudip Bhattacharyya</li><li>• Indian Academy of Sciences summer research fellow, <b>ISRO</b>, Bangalore (June–Aug 2015)<br/>Advisor: Dr. Anuj Nandi</li><li>• Visiting Student Program, <b>IITp</b>, Bangalore (May–Dec 2015)<br/>Advisor: Prof. Aruna Goswami</li></ul> |  |

1. **Sridhar, N.**, Sobacchi, E., Sironi, L., et al. 2026. “*Interaction of strong electromagnetic waves with unmagnetized pair plasmas*” arXiv:2604.11698; accepted in PRL.
2. **Sridhar, N.**, Ripperda, B., Sironi, L., et al. 2025. “*Bulk Motions in the Black Hole Jet Sheath as a Candidate for the Comptonizing Corona*” ApJ 979, 199
3. **Sridhar, N.**, Metzger, B. D., & Fang, K. 2024. “*High-Energy Neutrinos from Gamma-Ray-Faint Accretion-Powered Hypernebulæ*.” ApJ 960, 74
4. **Sridhar, N.**, Sironi, L., & Beloborodov, A. M. 2023. “*Comptonization by Reconnection Plasmoids in Black Hole Coronæ II: Electron-Ion Plasma*.” MNRAS 518, 1301-1315
5. **Sridhar, N.**, & Metzger, B. D. 2022. “*Radio Nebulæ from Hyper-Accreting X-ray Binaries as Common Envelope Precursors and Persistent Counterparts of FRBs*” ApJ 937, 5
6. **Sridhar, N.**, Sironi, L., & Beloborodov, A. M. 2021. “*Comptonization by Reconnection Plasmoids in Black Hole Coronæ I: Magnetically Dominated Pair Plasma*” MNRAS 507, 5625-5640
7. **Sridhar, N.**, Metzger, B. D., Beniamini, Paz., et al. 2021. “*Periodic Fast Radio Bursts from Luminous X-ray Binaries*.” ApJ 917, 13
8. **Sridhar, N.**, Zrake, J., Metzger, B. D., et al. 2021. “*Shock-powered radio precursors of neutron star mergers from accelerating relativistic binary winds*.” MNRAS 501, 3184-3202
9. **Sridhar, N.**, García, J. A., Steiner, J. F., et al. 2020. “*Evolution of the Accretion Disk—Corona during the Bright Hard-to-soft State Transition: A Reflection Spectroscopic Study with GX 339-4*.” ApJ 890, 53
10. **Sridhar, N.**, Bhattacharyya, S., Chandra, S., & Antia, H. M. 2019. “*Broad-band reflection spectroscopy of MAXI J1535-571 using AstroSat: estimation of black hole mass and spin*.” MNRAS 487, 4221-4229

$\leq 3^{\text{rd}}$  author, with significant contributions (10): [ $\dagger$  = supervised student]

11. Uno, K. $\dagger$ , Metzger, B. D., **Sridhar, N.** et al. 2025. “*The Population of Radio Hypernebulæ from Mass-Transferring Massive Binaries*” to be submitted to ApJ
12. Gupta, S. $\dagger$ , **Sridhar, N.**, Sironi, L. 2023. “*Comptonization by Reconnection Plasmoids in Black Hole Coronæ III: Dependence on the Guide Field in Pair Plasma*.” MNRAS 527, 6065-6075
13. Gupta, S. $\dagger$ , **Sridhar, N.**, Sironi, L. 2025. “*The Role of Electric Dominance for Particle Injection in Relativistic Reconnection*” MNRAS 538, 49-59
14. Bhandari, S., Marcote, B., **Sridhar, N.**, et al. 2023. “*Constraints on the persistent radio source associated with FRB 20190520B using the European VLBI Network*”, ApJ 958, 2
15. Eftekhari, T., Fong, W-F., Gordon, A. C., **Sridhar, N.**, et al. 2023. “*An X-ray census of Fast Radio Burst host galaxies: constraints on AGN and X-ray counterparts*”, ApJ 958,1
16. Metzger, B. D., **Sridhar, N.**, Margalit, B., et al. 2022. “*A Toy Model for the Time-Frequency Structure of Fast Radio Bursts: Implications for the CHIME/FRB Burst Dichotomy*.” ApJ 925, 135
17. Margalit, B., Beniamini, P., **Sridhar, N.**, Metzger, B. D., 2020. “*Implications of a Fast Radio Burst from a Galactic Magnetar*.” ApJL 899, L27
18. García, J. A., Tomsick, J. A., **Sridhar, N.**, et al. 2019. “*The 2017 Failed Outburst of GX 339-4: Relativistic X-Ray Reflection near the Black Hole Revealed by NuSTAR and Swift Spectroscopy*.” ApJ 885, 48
19. Bhattacharyya, S., Yadav, J. S., **Sridhar, N.**, et al. 2018. “*Effects of Thermonuclear X-Ray Bursts on Non-burst Emissions in the Soft State of 4U 1728-34*.” ApJ 860, 88
20. Karinkuzhi, D., Goswami, A., **Sridhar, N.**, et al. 2018. “*Chemical analysis of three barium stars: HD 51959, HD 88035, and HD 121447*.” MNRAS 476, 3086-3096

$> 3^{\text{rd}}$  author (16):

21. Shuvo, P., Meyer, E., ..., **Sridhar, N.**, et al. 2026. “*First Detection of Radio Polarization During Jet Formation in the Changing-Look AGN 1ES 1927+654*” arXiv:2700.00000; Submitted to ApJ.
22. Uttarkar, P., Shannon, R., ..., **Sridhar, N.**, et al. 2026. “*A fast radio burst cyclone in technicolour: evidence of plasma lensing*” arXiv:2602.16409; Submitted to Nature.

23. Moroianu, A., Bhandari, S., ..., **Sridhar, N.**, et al. 2026. “*A Milliarcsecond Localization Associates FRB 20190417A with a Compact Persistent Radio Source and an Extreme Magnetoionic Environment*” *ApJL* 996, L16
24. Snelders, M. P., Hessels, J. W. T., ..., **Sridhar, N.**, et al. 2025. “*Revisiting FRB 20121102A: milliarcsecond localisation and a decreasing dispersion measure*” *arXiv:2510.11352*
25. Blanchard, P., Berger, E., ..., **Sridhar, N.**, et al. 2025. “*James Webb Space Telescope Observations of the Nearby and Precisely Localized FRB20250316A: A Potential Near-IR Counterpart and Implications for the Progenitors of Fast Radio Bursts*” *ApJL* 989, L49
26. Ibik, A., Drout, M., ..., **Sridhar, N.**, et al. 2024. “*A search for persistent radio sources toward repeating fast radio bursts discovered by CHIME/FRB*” *ApJ* 976, 199
27. Dong, Y., Eftekhari, T., ..., **Sridhar, N.**, et al. 2024. “*A Radio Study of Persistent Radio Sources in Nearby Dwarf Galaxies: Implications for Fast Radio Bursts*”, *ApJ* 973, 133
28. Sarin, N., Clarke, T. A., ..., **Sridhar, N.** 2024. “*The origin of the coherent radio flash potentially associated with GRB 201006A*”, *ApJL* 973, L20
29. AXIS TDAMM SWG., ..., incl. **Sridhar, N.**, et al. 2023. “*Prospects for Time-Domain and Multi-Messenger Science with AXIS*”, *Universe*, 10, 316
30. Connors, R. M. T., Tomsick, J. T., ..., **Sridhar, N.**, et al. 2023. “*The High Energy X-ray Probe (HEX-P): Probing Accretion onto Stellar Mass Black Holes*”, *Front. Astron. Space Sci.* 10:1292682
31. Kammoun, E., Lohfink, A. M., ..., **Sridhar, N.**, et al. 2023. “*The High Energy X-ray Probe (HEX-P): Probing the physics of the X-ray corona in active galactic nuclei*”, *Front. Astron. Space Sci.* 10:1308056
32. Dong, Y., Eftekhari, T., ..., **Sridhar, N.**, et al. 2023. “*Mapping Obscured Star Formation in the Host Galaxy of FRB 20201124A*”, *ApJ* 961, 44
33. Bhandari, S., Gordon, A. C., ..., **Sridhar, N.**, et al. 2023. “*A non-repeating fast radio burst in a dwarf host galaxy*”, *ApJ* 948, 67
34. Connors, R. M. T., García, J. A., ..., **Sridhar, N.**, et al. 2021. “*Reflection Modeling of Black Hole Binary 4U 1630–47: Disk Density and Returning Radiation*”, *ApJ* 909, 146
35. Connors, R. M. T., García, J. A., ..., **Sridhar, N.**, et al. 2020. “*Evidence for Returning Disk Radiation in the Black Hole X-Ray Binary XTE J1550–564.*” *ApJ* 892, 47
36. Connors, R. M. T., García, J. A., ..., **Sridhar, N.**, et al. 2019. “*Conflicting Disk Inclination Estimates for the Black Hole X-Ray Binary XTE J1550–564.*” *ApJ* 882, 179
37. Verdhun Chauhan, J., Yadav, J. S., ..., **Sridhar, N.**, et al. 2017. “*AstroSat/LAXPC Detection of Millisecond Phenomena in 4U 1728–34.*” *ApJ* 841, 41

Books and white papers (2):

38. Koss, M., ., incl. **Sridhar, N.**, et al. 2025. “*The Advanced X-ray Imaging Satellite Community Science Book*”, *arXiv:2511.00253*
39. Safi-Harb, S., Burdge, K., ..., incl. **Sridhar, N.**, et al. 2023. “*From Stellar Death to Cosmic Revelations: Zooming in on Compact Objects, Relativistic Outflows and Supernova Remnants with AXIS*”, *arXiv:2311.07673*

Astronomical Telegrams (2):

40. Blanchard, P., Berger, E., ..., **Sridhar, N.**, et al. 2015. “*Deep JWST/NIRCam Imaging of FRB 20250316A: Detection of Potential IR Counterparts.*” *ATel* #17195
41. García, J. A., Harrison, F., ..., **Sridhar, N.**, et al. 2017. “*NuSTAR Observation of GX 339–4 in the early stages of its 2017 outburst.*” *ATel* #10825

OBSERVING X-ray: 15 Ms; Radio: 142 ks; Optical: 65 ks; IR: 4.5 hr  
 EXPERIENCE Total grant: \$624,935  
As Principal Investigator:

1. NICER archival data of transitional ms pulsar (Cycle AO8; 1.2 Ms of data; US\$45,000).
2. Joint NICER and NuSTAR observations studying reflection features from X-ray binaries (Cycle AO3; 60 ks; US\$37,500).
3. uGMRT multi-band radio observations of Giant Radio Pulses (Cycle 34; 36 ks).
4. *AstroSat* observations of ~10 sources: accretion powered millisecond pulsars, neutron star type-I bursts, black hole X-ray binaries (Cycles A04-A08; cumulatively >300 ks).

As Co-Investigator:

1. JWST search for a transient NIR counterpart to FRB 20250316A (PI: Peter Blanchard; \$110,500 ; cycle 5, 2026).
2. JWST observation of FRB 20250316A (PI: Peter Blanchard; cycle 4, 2025).
3. Joint NuSTAR and NICER observation to study the reflection properties and evolution of accretion disks in black hole binaries (Cycle 10; 500 ks; PI: J. Garcia).
4. GBT observations to probe the dynamic environment of FRB 20180301A (Cycle 2024B; 108 ks; PI: P. Kumar).
5. Chandra observations to probe the engines of fast radio bursts (Cycle 25; 106 ks; PI: T. Eftekhari).
6. GBT observations to probe the local environment of FRB 20180301A (Cycle 2023B; 72 ks; PI: P. Kumar).
7. VLA observations to constrain FRB ‘persistent radio sources’ (Cycle 2023B; 10 ks; PI: Y. Dong).
8. NICER observations for amassing a comprehensive spectral timing archive of outbursting black holes (Cycle AO5; 200 ks; \$43,000; PI: J. Steiner).
9. Joint NICER and NuSTAR observation to study the reflection properties and evolution of accretion disks in black hole binaries (Cycle AO4; 110 ks; \$44,000 ; PI: J. Garcia).
10. Joint NICER and NuSTAR observation of XRBs during its outburst (Cycle AO4; 88 ks; \$44,000; PI: R. Connors).
11. NICER observations for black hole spin measurements using continuum fitting (Cycle AO4; 40 ks; \$37,500; PI: J. Steiner).
12. Joint NICER and NuSTAR observations for measuring black hole spin and mass through X-ray reflection spectra and reverberation lags (Cycle AO4; 30 ks; PI: G. Mastroserio).
13. Joint XMM-Newton and NuSTAR observations for studying truncated accretion disks in black holes (AO 21; 120 ks; \$50,656; PI: J. García).
14. Chandra observation of nearby Fast Radio Bursts (Cycle 23; 55 ks; \$52,519; PI: T. Eftekhari).
15. Joint NICER and NuSTAR observation of XRBs during its outburst (Cycle AO3; 88 ks; \$37,500; PI: R. Connors).
16. NICER observations for black hole spin measurements using continuum fitting (Cycle AO3; 40 ks; \$44,000; PI: J. Steiner).
17. Joint NICER and NuSTAR observations for measuring black hole spin and mass through X-ray reflection spectra and reverberation lags (Cycle AO3; 30 ks; \$37,500; PI: G. Mastroserio).
18. NICER observations monitoring the evolving thermal disk emission of LMC X-3 (Cycle AO2; 60 ks; \$20,229; PI: J. Steiner).
19. Joint NICER and NuSTAR observation of XRBs during its outburst (Cycle AO2; 90 ks; \$21,031; PI: R. Connors).

Others:

1. Mt. Palomar 200” Hale telescope: DBSP observations of 14 variable AGN (2 nights)
2. NRAO Green Bank Telescope: radio pulsar observations (24 ks)

DELIVERED

TALKS/

PRESENTATIONS

(\*29 INVITED)

83. Talk and session chair at the FRB 2026 conference, **Guiyang**, China (July 2026)
82. \*Seminar at Department of Astronomy, **Tsinghua University**, Beijing (July 2026)
81. Talk at the BlackHolic 2026 conference, **Oxford University**, UK (March 2026)
80. \*Astrophysics seminar, **Washington University**, St. Louis, MO (Oct 2025)
79. 22nd AAS-High Energy Astrophysics Division meeting, St. Louis, MO (Oct 2025)
78. \*Black-board talk, Frontiers of Relativistic Plasma Physics, **KITP**, USA (Sep 2025)
77. \*Overview of FRB emission mechanisms, FRB 2025 conference, **McGill**, Canada (July 2025)
76. Seminar, **Inter-University Center for Astronomy & Astrophysics**, Pune (June 2025)
75. \*Astrophysics colloquium, **Tata Institute of Fundamental Research**, Mumbai (May 2025)
74. Astrophysics and Relativity seminar, **International Center for Theoretical Sciences**, Bangalore (May 2025)
73. Relativistic Magnetospheres workshop, **Ecole De Physique Des Houches**, France (April 2025)
72. Magnetic fields around Compact Objects workshop, **Perimeter Institute**, Canada (March 2025)
71. Fast Radio Burst Frontiers workshop, **Carnegie Mellon University**, USA (March 2025)
70. \*Talk at the XOC group meeting **Stanford**, CA (Sept 2024)

69. Talk at the High-energy plasma phenomena in astrophysics workshop **MIAPbP, Munich, Germany** (Sept 2024)
68. Graduating student colloquium, **Columbia University, NY** (April 2024)
67. \***AXIS** seminar series (April 2024)
66. 21st AAS-High Energy Astrophysics Division meeting, Texas (April 2024)
65. \*Seminar, Simons Collaboration on Extreme Electrodynamics of Compact Object Sources (**SCEECs**) (Feb 2024)
64. \*Astroplasmas seminar, **Princeton University, NJ** (Dec 2023)
63. \*CIERA Theory seminar, **Northwestern University, IL** (Nov 2023)
62. JSI workshop on ‘Winds throughout the Universe’, **Annapolis, MD** (Oct 2023)
61. \*Theoretical astrophysics group board talk, **Carnegie Observatories, CA** (Sept 2023)
60. \*Seminar, **Syracuse University, NY** (Sept 2023)
59. \*Review talk at Astrophysics of Fast Radio Bursts II workshop (+session chair), **CCA, Flatiron Institute, NY** (Sept 2023)
58. \*Colloquium, **National Center for Radio Astrophysics, Pune, India** (June 2023)
57. \*Astrophysics seminar, **Tata Institute of Fundamental Research, Mumbai**(May 2023)
56. \*‘Transient Tuesday’ talk, DARK, Niels Bohr Institute **Copenhagen, Denmark**(May 2023)
55. \*Discussion chair, Fast Radio Burst Follow-Up workshop, **University of Toronto** (April 2023)
54. \*High energy astrophysics seminar, **Hebrew University of Jerusalem, Israel**(March 2023)
53. \*Seminar, Astrophysics Research Center of **The Open University, Israel** (March 2023)
52. \*High energy astrophysics meeting, Black Hole Initiative, **Harvard, MA** (Feb 2023)
51. Cosmic transients lab seminar, Center for Astrophysics, **Harvard, MA** (Feb 2023)
50. High Energy Astrophysics seminar, Center for Astrophysics, **Harvard, MA** (Feb 2023)
49. FRB Taiwan 2023, National Chung Hsing University, Taichung **Taiwan** (Feb 2023)
48. \*Workshop on ‘Overcoming disconnects in the understanding of accreting black holes’, **Lorentz Center, Leiden, The Netherlands** (Jan 2023)
47. HEX-P black hole binaries, **science working group** presentation (Dec 2022)
46. \*CCAPP seminar, **Ohio State University, OH** (Nov 2022)
45. 64th APS Division of Plasma Physics annual meeting, **Spokane, WA** (Oct 2022)
44. Review talk on Observations of FRBs, **CCA, Flatiron Institute, NY** (Oct 2022)
43. \***AXIS** ‘compact objects and supernova remnants’ **science working group** (Oct 2022)
42. \*Astrophysics seminar, **Cornell University, NY** (Sept 2022)
41. \*AGN seminar at **NASA** Goddard Space Flight Center, MD (Sept 2022)
40. **NASA** Time Domain and Multi-Messenger (TDAMM) Initiative Workshop, **Annapolis, MD** (Aug 2022)
39. FRB 2022 meeting, IAUGA, **Busan, South Korea** (Aug 2022)
38. Summer BLAST workshop, Black Hole Initiative, **Harvard, MA** (July 2021)
37. **AXIS** ‘Time-Domain and Multi-Messenger Astronomy’ **science working group** (July 2022)
36. Special pizza lunch seminar, **Caltech, CA** (June 2022)
35. Seminar at **Carnegie Observatories, CA** (June 2022)
34. 240th meeting of the American Astronomical Society, **Pasadena, CA** (June 2022)
33. Workshop on Relativistic Plasma Astrophysics, **Purdue University, IN** (May 2022)
32. Compact objects group meeting at **CCA, Flatiron Institute, NY** (April 2022)
31. \*Kapteyn Astronomical Institute, **Groningen, The Netherlands** (April 2022)
30. 19th AAS-High Energy Astrophysics Division meeting, **Pittsburgh, PA** (March 2022)
29. Brown bag lunch seminar, **Massachusetts Institute of Technology, MA** (Feb 2022)
28. Winter BLAST workshop, Black Hole Initiative, **Harvard, MA** (Dec 2021)
27. Gothamfest, **CCA, Flatiron Institute, NY** (Dec 2021)
26. Kavli-IPMU workshop on particle acceleration by reconnection, **Tokyo, Japan** (Nov 2021)
25. 9th Microquasar workshop, **Cagliari, Italy** (Sept 2021)
24. Astrofest, **Columbia University, NY** (Sept 2021)
23. High-energy plasma phenomena in astrophysics, **Munich, Germany** (July 2021)
22. FRB 2021 conference (online) (July-Aug 2021)
21. \*Princeton astro-coffee, **Princeton, NJ** (July 2021)
20. ‘Midwest Magnetic Fields’ Workshop, **Wisconsin-Madison,** (June 2021)
19. \*CHIME/FRB collaboration, **McGill, Canada** (May 2021)
18. \*Seminar at the **Institute of Astrophysics - FORTH, Greece** (April 2021)
17. Compact objects group meeting at **CCA, Flatiron Institute, NY** (March 2021)
16. **Kyoto** FRB meeting, **Yukawa Institute for Theoretical Physics** (Feb 2021)

15. Compact object group meeting, **TIFR**, Mumbai (Dec 2020)
14. 62nd Annual Meeting of the **APS Division of Plasma Physics** (Nov 2020)
13. Compact objects group meeting, **CCA, Flatiron Institute**, NY (June, Oct, Nov 2020)
12. Plasma astrophysics summer program, **CCA, Flatiron Institute**, NY (July 2019)
11. Theoretical High Energy Astrophysics seminar, **Columbia University** (June 2019)
10. Graduate Seminar, **Columbia University**, NY (Feb, April 2019)
9. Research Seminar, **Columbia University**, NY (Dec 2018)
8. Pizza lunch talk, **Columbia University**, NY (Oct 2018, 2022)
7. *NuSTAR* SOC group meeting, **Caltech**, CA (Sept 2017)
6. Group presentation at **NRAO Green Bank Observatory**, WV (Sept 2017)
5. 16th AAS-High Energy Astrophysics Division meeting, **Sun Valley**, ID (March 2022)
4. Visiting Student Program ceremony, **University of Alberta**, Canada (July 2017)
3. Lunch talk at **University of Alberta**, Canada (June 2017)
2. ‘Wide band X-ray spectro-temporal studies’, **TIFR**, Mumbai, India (Jan 2017)
1. **IISER Bhopal** Physics department symposium (Nov 2015)

MENTORING Total: 9 students

- *Ningyue Fan*; **Ph.D.** student, Stanford University (Fall 2025–)
  - *Christopher Farias*; **undergraduate** student, San Diego State University (Summer 2025)
  - *Kohki Uno*; **Ph.D.** student, Kyoto University (1 paper) → postdoc at Columbia University (Summer 2024)
  - *Arnab Kundu*; **masters** thesis student, IISER Kolkata (2021–2022)
  - *Akash Vani*; Pune University → Masters at Heidelberg University → Ph.D. student at MPA, Garching (2019–2020)
- At Columbia University:
- *Sanya Gupta*; **undergraduate** research projects (2 papers) → Ph.D. student at Stanford University (2022–2025)
  - *Shifra Mandel*; **graduate** mentor-mentee program (2021–2022)
  - *Shuhan Zhang*; Women in Science at Columbia **undergraduate** mentorship program → Ph.D. student at UC San Diego (2021)
  - *Bhairavi Chandrasekhar*; **undergraduate** summer internship (2020)

TEACHING At Columbia University:

- **Head TA** of the Astronomy department (2020–2021)
- **Instructor** for undergraduate Intro to Astronomy lab courses (Fall 2019, Spring 2020)
- **Teaching Assistant** for 2 undergraduate (non-major) courses (Fall 2018, Spring 2019)
- **Guest lectured** for ASTR UN1403 (in lieu of Prof. David Helfand) (Feb 2020)
- **Advanced track** in Columbia CTL’s [Teaching Development Program \(TDP\)](#) (2018–2024)
- Completed CTL’s [Innovative Course Design Seminar](#) (2021)

Outside Columbia University:

- Selected as a **Lead instructor** at the Yale Young Global Scholars (YYGS) program **Yale University**, New Haven, CT (Summer 2020; postponed due to Covid-19)
- Trained to incorporate **inquiry-based learning techniques** in STEM education through the [ISEE—Professional Development Program \(PDP\)](#), **UC Santa Cruz** (March–Sept 2019)
- [Designed and taught an inquiry activity](#) on statistics, **UC Los Angeles** (Sept 2019)

- PROFESSIONAL LEADERSHIP
- Scientific Organizing Committee of symposium on *The birth, life and death of black holes*, European Astronomical Society annual meeting, Leiden, The Netherlands (June 2021)
- At Stanford University:
- Postbaccalaureate program admissions committee member (2025)
  - Ph.D. program admissions committee member (2024–2025)
  - Started the weekly KIPAC Compact Objects Group meetings (Oct 2024–)
- At Columbia University:
- Organized an eight-part lecture series titled *‘Science Policy for Scientists’* (Spring 2023)
  - Department representative, Arts and Sciences Graduate Council (2022–2023)
  - Graduate representative, faculty hiring committee, Astronomy department (2021–2022)
  - Graduate representative, colloquium committee, Astronomy department (2022–2023)
  - Graduate admissions committee member, Astronomy department (2021–2022)
  - Co-convenor of *Astrofest* (Sept 2020)
- At IISER Bhopal:



|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | <ul style="list-style-type: none"> <li>• Senate representative, Central Advisory Committee, Students Activity Council (2016-2017)</li> <li>• Elected Science Council secretary (2014-2016)</li> <li>• Convenor of <i>Singularity</i>, the Institute annual science festival (April 2015)</li> <li>• Founder of the Institute <i>Astronomy Club</i> (2014)</li> <li>• Founder of the Institute <i>Quiz Club</i> (2014)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| PROFESSIONAL SERVICE           | <ul style="list-style-type: none"> <li>• Invited to serve as a <b>referee</b> for: <ul style="list-style-type: none"> <li>–Journal of Plasma Physics,</li> <li>–Proceedings of the National Academy of Sciences of the USA,</li> <li>–Nature,</li> <li>–The Astrophysical Journal,</li> <li>–Monthly Notices of Royal Astronomical Society,</li> <li>–Physical Review X,</li> <li>–Astronomy &amp; Astrophysics.</li> </ul> </li> <li>• Invited to serve as a <b>panelist</b> for: <ul style="list-style-type: none"> <li>–NASA’s Chandra, NICER observatories</li> <li>–National Science Foundation,</li> <li>–<i>AstroSat</i>,</li> <li>–NCRA-uGMRT,</li> <li>–Observer at Hubble Space Telescope time allocation committee.</li> </ul> </li> <li>• <b>Science working group member of:</b> <ul style="list-style-type: none"> <li>–NewAthena (2025–)</li> <li>–Advanced X-ray Imaging Satellite (<b>AXIS</b>) (2022–)</li> <li>–High-Energy X-ray Probe (<b>HEX-P</b>) (2022–)</li> <li>–Probe for far-Infrared Mission for Astrophysics (<b>PRIMA</b>) (2025–)</li> </ul> </li> <li>• DEI committee member, Columbia Arts and Sciences Graduate Council (2022-2023)</li> <li>• <b>Judge</b>, Chambliss poster competition, American Astronomical Society meeting (2022)</li> <li>• <b>Webmaster:</b> (1) <a href="#">Columbia Astronomy Department</a>, (2) <a href="#">Theoretical High-Energy Astrophysics group, Columbia</a>, (3) <a href="#">Simons Collaboration on Extreme Electrodynamics of Compact Sources</a></li> <li>• Member of American Astronomical Society (2018–present)</li> <li>• Member of New York Academy of Sciences (2018–2023)</li> </ul> |
| SUMMER SCHOOLS                 | <ul style="list-style-type: none"> <li>• <b>Multiscale modeling of astrophysical and space plasmas – 2019</b><br/>Center for Computer Astrophysics (CCA), Flatiron Institute, New York, NY</li> <li>• <b>Multiple approaches to plasma physics from laboratory to astrophysics – 2019</b><br/>Ecole de Physique Des Houches, Les Houches, France</li> <li>• <b>NRAO Green Bank Telescope observer training workshop – Fall 2017</b><br/>Green Bank Observatory, West Virginia</li> <li>• <b>Summer School on Gravitational Wave Astronomy – 2016</b><br/>International Center for Theoretical Sciences (ICTS), TIFR, Bangalore</li> <li>• <b>Summer school on relativity, black holes and neutron stars – 2015</b><br/>Indian Institute of Astrophysics (IIAp), Bangalore</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| COMPUTING/ SOFTWARE EXPERIENCE | <ul style="list-style-type: none"> <li>• Languages and tools: Python, git, FORTRAN, IRAF, gnuplot, DS9, L<sup>A</sup>T<sub>E</sub>X, <i>f</i>tools (HEASoft): XSPEC, QDP, XRONOS</li> <li>• Supercomputers: NASA Pleiades, DoE-NERSC Cori, NSERC WestGrid, Flatiron Institute HPC cluster, Columbia University Habanero, Terremoto, Ginsburg clusters</li> <li>• <u>Numerical tools</u>: Particle-In-Cell, Magnetohydrodynamical simulations</li> <li>• Telescope data handling: <i>AstroSat</i>, <i>Swift</i>, <i>RXTE</i>/PCA, <i>NuSTAR</i>, <i>NICER</i>, <i>XMM</i>-Newton, NCRA-GMRT, NRAO-GBT (Astrid, CLEO)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| SCIENCE OUTREACH               | <ol style="list-style-type: none"> <li>1. Public outreach talk during KIPAC community day, Stanford University (April 2025)</li> <li>2. Research career in astrophysics for undergraduates at IISER Bhopal (Jan 2025)</li> <li>3. <a href="#">KIPAC public outreach talk</a>, Stanford University (Dec 2024)</li> <li>4. Co-organized <a href="#">astronomy outreach activities</a> at Columbia University for the South African academic achievers program (750 high-school students) (Oct 2022)</li> <li>5. Talk on introduction/orientation to summer research, for undergraduate researchers at Columbia University (June 2022)</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

6. Popular talk on black hole X-ray binaries at IISER Bhopal Astronomy club (Oct 2021)
7. [Mentored the students of Democracy Prep High School](#), Harlem, NYC, on the foundations of computer coding (Fall 2021)
8. Research career in astrophysics for undergraduates at IISER Bhopal (April 2021)
9. Delivered a lecture on science with space-based X-ray telescopes, for high-school students at *Frontiers of Science*—the annual outreach program at TIFR, Mumbai. (Nov 2017)
10. Organized a tour of GMRT for TIFR–Visiting Summer Research students (June 2016)

News coverage and appearances:

- *Super sharp images with EVN reveal another Fast Radio Burst Coming from a Hypernebula.* [[JIVE](#); 29-Nov-2023][[ASTRON](#); 01-Dec-2023][[Universe Today](#); 05-Dec-2023]
- *Astronomers caught a potent radio burst blasting at us from a dwarf galaxy 3 billion light-years away* [**Popular Science**; 09-June-2022]
- *AstroSat Catches Nuclear Reactions Spreading Across a Neutron Star* [**The Wire**; 07-October-2021]
- *Long burst short: an unexpected supernova* [**Research Matters**; 26-July-2021]
- *Fast Radio Bursts from globular clusters and what could power them* [**National Geographic**; 27-May-2021]
- *Digging the grave: In search of the remains of merging black holes and neutron stars* [**Research Matters**; 14-Sept-2021]
- *FRBs from Galactic magnetar* [**The Wire**; 03-June-2020] [AAS Nova; 09-Sept-2020]
- *Black hole bends light on itself* [**Gizmodo**; 10-April-2020] [**Caltech News**; 08-April-2020] [**SYFY**; 09-July-2020]